Pre-Class Assignment: Chapter 4

Substitutes for Prediction

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

Reading: Gans, Chapter 4 (Substitutes for Prediction)

Instructions: Answer each question in 1–3 sentences. These questions are designed to be straightforward if you did the reading.

You must submit this assignment in person. The assignment must be handwritten. No photocopies will be accepted.

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1. Gans defines a **rule** and a **decision** in Definition 4.1. In your own words, what is the difference between the two?
2. In Section 4.1.1, why is it optimal for an agent following a rule to choose $a_i = 1$ when $p > \frac{1}{2}$?
3. When information processing costs c are involved, the agent chooses a decision over a rule if $e - p \geq \frac{c}{R-r}$. What happens to the threshold for making a decision as the accuracy e of the signal increases?
4. In Section 4.2, Gans discusses two risk management activities: **insurance** and **protection**. In one sentence each, what does each one do?
5. Give one example of insurance and one example of protection from the reading (other than the Notre Dame and hedgerow examples).
6. In Proposition 4.1, Gans shows that when an agent moves from a rule to a decision, **protection** is always reduced (to zero). Why does protection become unnecessary once the agent is making state-contingent decisions?
7. Proposition 4.1 also says that **insurance** may either rise or fall when an agent switches from a rule to a decision. What determines the direction?
8. Corollary 4.1 states: "When (x, Δ) are endogenous, a necessary condition for making a decision to be preferred to a rule is that $e > p$." Why does this condition still hold even when risk management activities are available?
9. In Section 4.4, Gans introduces **asymmetric stakes**. What does it mean for a decision to have "high stakes" in this framework, and how does it affect the adoption of AI prediction?
10. In one sentence, state the key takeaway of Chapter 4.